SVM Assignment

1. Understanding the Dataset

(a) Generate a two-class dataset using **make_blobs()** from scikit-learn with 100 samples and two features.

(b) Plot the dataset using a scatter plot. Use different colors for the two classes.

(c) What do the two features and two colors represent?

2. Training a Linear SVM

(a) Split the dataset into training and testing sets (80% training, 20% testing).

(b) Train a linear SVM using **SVC(kernel="linear")**.

(c) Predict the test labels and calculate classification accuracy.

3. Decision Boundary and Support Vectors

(a) Plot the dataset along with the SVM decision boundary.

(b) Display the two margin lines and highlight the support vectors.

(c) What are support vectors? Why are they important for the decision boundary?

4. Understanding Hinge Loss

(a) For a sample with true label **y = +1**, calculate the hinge loss **max(0, 1 - yf(x))** for model scores **f(x) = -1, 0, 0.5, 1, 2**.

(b) Plot hinge loss against model score for values from -2 to 3 when y = +1.

(c) At what value of **yf(x)** does hinge loss become zero?

5. Effect of Regularization (C)

(a) Train linear SVM models using **C = 0.1, 1, and 100** on the same dataset.

(b) Plot the decision boundary and margins for each value of C.

(c) Compare the margin widths and number of support vectors. What changes when C increases?

6. Nonlinear Data and Kernels

(a) Generate a nonlinear two-class dataset using **make_moons()** with noise = 0.2.

(b) Train one SVM with a linear kernel and another with an RBF kernel.

(c) Plot both decision boundaries and compare their classification accuracies. Which kernel works better on this dataset?

7. Simple SVM Implementation

(a) Implement a function in Python that computes the average hinge loss for a list of true labels (each -1 or +1) and prediction scores.

(b) Test your function with **y = [1, -1, 1]** and **scores = [0.5, 0.2, 1.5]**. Print the loss for each sample and the average loss.

8. Final Comparison

(a) Create a table comparing your trained models using kernel type, C value, training accuracy, and testing accuracy.

(b) Write 3-4 sentences explaining what you learned about margins, hinge loss, and kernels.